

UKRAINIAN CATHOLIC UNIVERSITY

BACHELOR THESIS

Maximizing Profit on the Upwork Platform Through Quantitative and Qualitative Analysis and Natural Language Processing

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*A thesis submitted in fulfillment of the requirements
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Declaration of Authorship

I, Oksana MATSKIV, declare that this thesis titled, “Maximizing Profit on the Up-work Platform Through Quantitative and Qualitative Analysis and Natural Language Processing” and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Signed:

Date:

“Never get so busy making a living that you forget to make a life.”

Dolly Parton

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Maximizing Profit on the Upwork Platform Through Quantitative and Qualitative Analysis and Natural Language Processing

by Oksana MATSKIV

Abstract

Would you believe we live in a crisis worse than the Great Depression? You might not, but if you saw charts of gaps between incomes and prices of houses, colleges, food, etc., you would see why. For example, let's go back to 1930: median household income in the US was \$1.3k while median house prices stood at \$3.9k[2], resulting in a price-to-income ratio of 3. Even during the Subprime mortgage crisis in 2007, when incomes rose to \$50.2k and houses cost \$257.4k[6], this rate was 5.1. Fast forward to 2023: median household incomes hit \$67.5k[10] and houses soared to \$417.7k[4], resulting in a ratio of 6.2 and it still continues to rise.

With this uncertainty, more people are taking on additional jobs to increase their income: some turn to freelancing due to its benefits like flexible schedules and the possibility of combining it with a main job, and others rely on it as their main income. Today, 43% of Zoomers and 46% of Millennials engage in one-time projects, making up 39% of the total U.S. workforce[9], and by 2027 this number may rise up to 50.9%[7]. And the world's largest online workplace with over 18 million freelancers and 5 million clients is Upwork[3].

The objective of this thesis is to determine the key factors that define the project's budget by leveraging NLP, quantitative and qualitative analyses. The results of this analysis will provide insights into project pricing dynamics with corresponding prediction models.

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List of Abbreviations

CV	C ategorical V ariable
NV	N umeric V ariable
TV	T ext V ariable
NTL	N atural L anguage P rocessing
ML	M achine L earning
MAE	M ean A bsolute E rror
MSE	M ean S quared E rror
RMSE	R oot M ean S quared E rror
MAPE	M ean A bsolute P ercentage E rror
SMAPE	S ymmetric M ean A bsolute P ercentage E rror
SHAP	S Hapley A dditive e x P lanations
IQR	I nter Q uartile R ange

Dedicated to my deceased grandfather...

Chapter 1

Introduction

1.1 Motivation

Central to my motivation is my aim to help knowledge grow in this area. Starting my thesis work, I realized there was a lack of studies on the dependencies I was investigating. Over time some research appeared, but they still had gaps and my work aims to fill in the gaps. I will perform new research that will give meaningful input to increase our grasp of current job trends and offer direction for further explorations.

Today, 43% of Zoomers and 46% of Millennials engage in one-time projects, making up 39% of the total U.S. workforce[9], and by 2027 this number may rise up to 50.9%[7]. These numbers testify to the growing relevance of freelancing therefore, this thesis aims to explore all the nuances of freelancing to provide insights into navigating this evolving landscape of work.

As I often use platforms like Upwork, I want to truly understand how job hunting works inside out. Moreover, Upwork helped me grow a strong professional network that was involved in my research process through interviews and feedback, ensuring its veracity and versatility.

1.2 Why Upwork

Upwork is rapidly becoming a top contender against competitors like Fiverr, Freelancer, and Freelancehunt. The reasons below reflect why Upwork is best for my study.

Upwork hosts over 18 million freelancers and 5 million clients, which makes it the biggest online freelance marketplace[3]. With this user base and annual earnings exceeding \$2 billion as reported by VCBay[5], it showcases Upwork's assurance offered to its partners in the freelance work field.

Upwork's reach goes beyond its users: It has received various recognitions like a place on Time's TIME100 Most Powerful Companies in 2022 list[8]. This honor shows both the importance of Upwork in business and its key part in shaping work and enterprise's future.

Upwork mainly supports long-term projects, unlike many platforms that favor brief ones. This approach promotes steadiness and constant flow for workers and customers alike and helps maintain lasting work connections for ongoing teamwork.

The method Upwork employs differs from the way Fiverr operates, where job posts can be seen only by those who receive invites. On Upwork, employers post job posts with full details, including key factors needed for in-depth review.

1.3 Problem Statement, Objectives, and Expected Outcome

The main objective of this thesis is to maximize freelancers' profit on the Upwork platform that involves two approaches:

1. Improving freelancer profiles and portfolios, receiving positive feedback from previous clients, and writing cover letters with relevant case studies to attract potential customers.
2. Identifying and applying for jobs with factors indicating high profitability potential.

While both approaches are essential, this thesis focuses on the second part, as cover letters with case studies attached are only visible to employers who have been directly contacted by freelancers.

The main problems regarding choosing jobs are:

1. Freelancers may experience difficulty finding good jobs due to ignorance over what factors should be considered.
2. Certain employers might put unrealistic rates or avoid specifying pricing details completely.

Therefore, the main goals of this thesis are to:

1. Define key factors that indicate the project's profitability
2. Create a complex prediction model that will predict the project budget considering all the factors.

1.4 Structure Of The Thesis

Below are outlined the remaining chapters of the thesis:

Chapter 2 provides a description of the dataset and its columns.

Chapter 3 explains theory and defines methodology.

Chapter 4 provides conducting interviews, exploring existing studies, and forming hypotheses.

Chapter 5 provides data reprocessing and quantitative analysis of categorical and numeric variables.

Chapter 6 provides results of models with the best and worst performance.

Chapter 7 defines summary and further work.

Chapter 2

Dataset Overview

The dataset prepared by Ahmed Myalo ("Upwork freelance jobs +60k") contains more than 60,000 job posts made within two months. Below is provided the list of column entries categorized as Text Variables (TV), Categorical Variables (CV), and Numeric Variables (NV) with a short description:

Job Title (TV) — Title of the job posting.

EX_level_demand (CV) — Desired expertise level ("Entry level", "Intermediate", or "Expert")

Search_Keyword (CV) — Keyword used to find the job ("Developer", "Marketing", "Data Science", or "3D")

Description (TV) — Full job description provided by the client.

Payment_Situation (CV) — Indicates if the client's payment card is activated and verified.

Enterprise_Client (CV) — Identifies whether the client is a big company or not.

Spent(\$) (NV) — Total amount spent by the client on Upwork.

Client_Country (CV) — Client's location.

Rating (NV) — Client's rating on Upwork.

Feedback_Num (NV) — Number of feedback from previous freelancers.

Payment_type (used only in preprocessing) — Indicates if a project has fixed price or is hourly-based.

Job_Cost (NV) — Project price for fixed-price job.

Hourly_Rate (NV) — Hourly rate paid for hourly-based job.

Chapter 3

Theoretical Background and Methodology

3.1 Qualitative Analysis

Qualitative analysis is a method of exploring that allows us to understand and interpret the data in the way numbers cannot. While quantitative analysis works mainly with statistical or ML approaches, qualitative analysis focuses on capturing the underlying meanings and patterns present in the data.

First, I will conduct a qualitative analysis involving interviews with experts in the field. By developing open-ended questions, I will identify which variables in my study are significant for further analysis. This will be followed by a review of relevant studies to identify existing research and knowledge gaps to further my understanding of the topic. In the end, hypotheses will be developed based on the findings from all previous steps to define a clear focus and direction for my research.

3.2 Data preprocessing

Preprocessing is the process of cleansing and transforming data to make it suitable for statistical or ML methods of data analysis. In this thesis we discrete next steps to avoid biases and make our research trustworthy and accurate:

Selecting only relevant data that have correlations we want to explore to reduce noise and improve model performance for better predictions.

Duplicate removal for data consistency, i.e. to ensure that each data point contributes only once to the analysis. Moreover, duplicates can lead to overfitting, where the model learns to memorize the training data rather than predict the testing data.

Handling missing data, as many ML algorithms fail if the dataset contains them. It can be performed by deleting rows with missing values completely if missing values represent over 30% or imputation of the missing values otherwise. Imputation is the method of handling missing data by replacing it with an estimated value, minimum, maximum, average, mode, or other.

Handling outliers, as they affect data distribution and may subsequently cause overfitting due to the model learning to capture noise rather than true underlying patterns. One common method to remove outliers is the 1.5xIQR rule which states that the value is an outlier if it is more than 1.5xIQR above the third quartile or below the first quartile. Important to note, that some outliers may be novelties, some exceptions of the general tendency but also entry or preprocessing errors.

For normalizing exclusively numerical data we can perform scaling by min-max scaling or z-score standardization, as many ML algorithms are sensitive to the scale of the input features. The difference between them is that min-max scales features to a specific range, typically between 0 and 1, while z-score transforms numerical variables so that they have a mean of 0 and a standard deviation of 1.

If data have clearly defined groups, we can perform clustering, i.e. divide similar data points into subgroups based on their patterns, for further processing of each cluster. The elbow method may be applied to define the optimized number of clusters.

With categorical data, we may perform feature creation and feature encoding.

Feature creation is the process of combining 2 or more features to create a new one that captures their interactions if they separately do not show essential correlation; Also, it may apply to reduce data dimensions.

As categorical data have a finite number of values, there is no need to apply NLP to every single word in each category, we can encode them using ordinal or nominal encoding. Ordinal encoding is applied, when there is a defined hierarchy among values (e.g. "small", "middle", "big") and we can map them to integers 1, 2, 3,... accordingly. On the other hand, nominal, i.e. one-hot or dummy encoding, is applied when there is no defined hierarchy among values (e.g. "red", "green", "blue") and is implemented by transforming variables to a binary variable 0 or 1 and creating corresponding data columns (e.g. "is_red", "is_green", "is_blue").

As computers operate only using numeric data, it is necessary to convert textual data into its multidimensional vector representation before it can be used for further analysis.

There are multiple ways to encode texts: token count representation (BoW - bag of words), capturing term importance by computing TF-IDF (Term Frequency-Inverse Document Frequency), dense vector representations that capture the semantic information (Word2Vec, GloVe, etc.), or encoders that capture contextual information (BERT).

Converting text in the lower- or uppercase simultaneously, as words like "HEART" and "heart" will be encoded differently, although it is the same word.

URL pattern removal, as they contain random letters and symbols that do not contribute to the semantics of the text. Punctuation removal, as those symbols also do not contribute to the semantics of the text, and words like "heart" and "heart," will be encoded differently. Numbers removal, as it allows to focus on the qualitative aspect of the statement rather than the specific quantity.

After text standardization, we can move on to stop words removal, as they frequently appear in the text, but typically carry little to no semantic meaning (e.g. "are", "and", "the").

The last step performed may be stemming or lemmatization. While stemming is a rule-based technique that blindly removes words endings to extract their stem (e.g. "loved", "love", "loving" -> lov), lemmatization is a more sophisticated technique that reduces words to their canonical forms, known as lemmas (e.g. "is", "are", "am" -> "be").

3.3 Quantitative Analysis

After conducting qualitative analysis and performing data preprocessing, we can scrutinize numerical attributes to explore patterns and correlations that underlie the data's essence.

Before creating any regressions, it is necessary to calculate count, min, max, mean, median, and mode to check if these characteristics will vary for every selected variable. We also may create a heat map that will graphically show if there are any correlations between variables, their direction, and strength.

The direction of correlation can be also calculated by Pearson's correlation coefficient. It indicates the degree of linear association between two continuous variables and has a range from -1 to +1, where a value close to +1 indicates a strong positive

correlation, a value close to -1 indicates a strong negative correlation and a value near 0 means little to no correlation.

An analog to the previous coefficient is Spearman's Rank Correlation Coefficient which evaluates the strength and direction of association between two variables based on the ranks of their values rather than their actual numerical values. This method captures monotonic relationships, which may not necessarily be linear, and is also less sensitive to outliers

After we explored and verified existing correlations we can test previously made hypotheses to validate the reliability and reproducibility of this thesis and move further with creating corresponding prediction models.

3.4 Building Prediction Models

After data preprocessing and correlations exploring, we can now proceed to building predicting models, as ML is a comprehensive tool that combines quantitative and NLP analyses.

There are different types of prediction models, each with its advantages, limitations, and applications, but this thesis mainly focuses on tree-based models, as they capture non-linear relationships by organizing data into a tree structure, and are more suitable for our data.

Basing on the article 'Understanding Tree-Based Models: A Simple Guide' by Dr. Stylianos (Stelios) Kampakis (2023)[1], we can describe the following tree-based models along with their distinctive features:

Decision trees are versatile models used for classification and regression tasks when a complex relationship exists between features and the target variable.

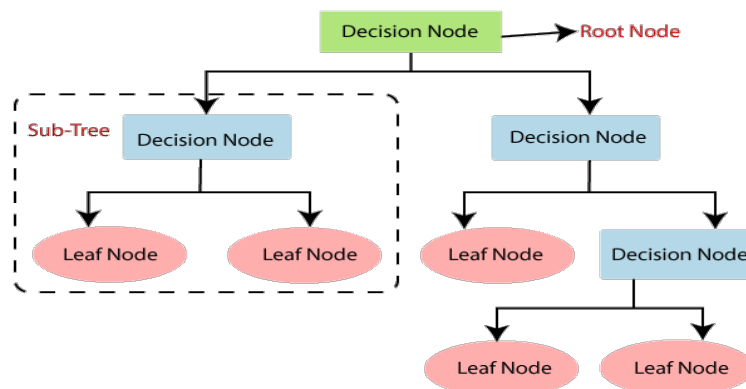


FIGURE 3.1: Decision tree algorithm example

Ensembling is a process of enhancing model performance by merging multiple models into a complex one. These methods include bagging, boosting, and stacking.

Bagging improves the accuracy of a model by merging multiple instances of the same algorithm and training them on random subsets of the training data.

Random forest is an advanced version of bagging that is specifically applied to decision tree models. It creates many decision trees in the training phase and each tree is trained on a unique portion of the data. Furthermore, during tree division, only a random subset of features is analyzed. Finally, all trees are combined using a basic voting system to make the ultimate prediction.

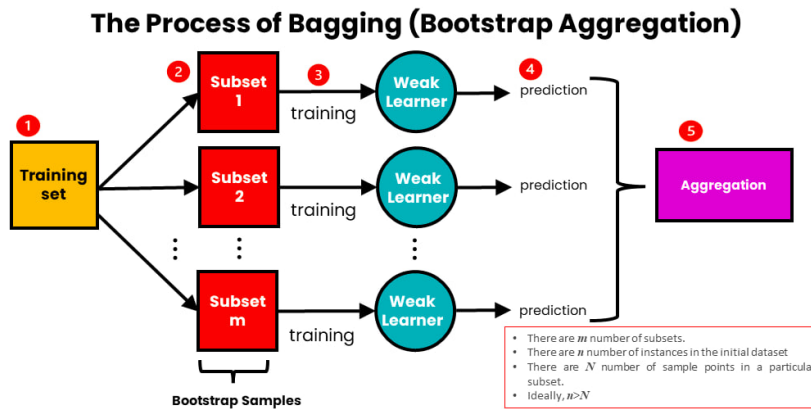


FIGURE 3.2: Bagging algorithm example

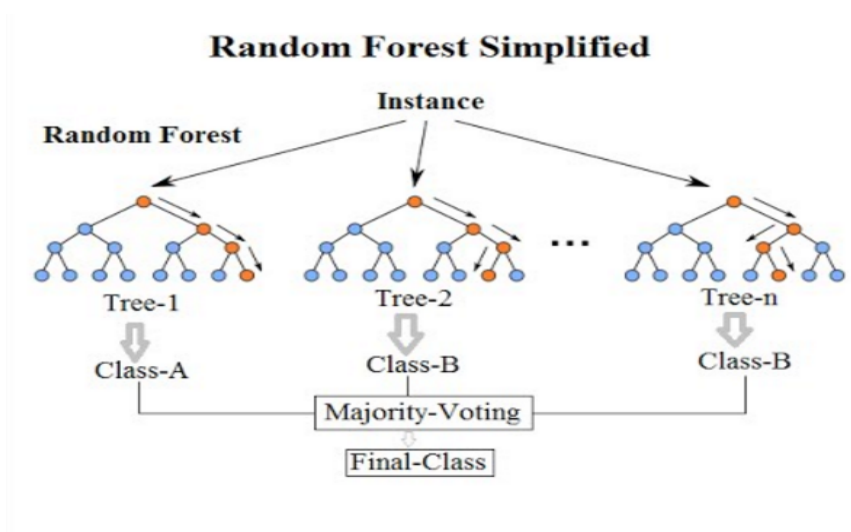


FIGURE 3.3: Random forest algorithm example

Unlike bagging, where models are trained independently, gradient boosting builds models sequentially, with each new model focusing on correcting the errors made by the previous ones.

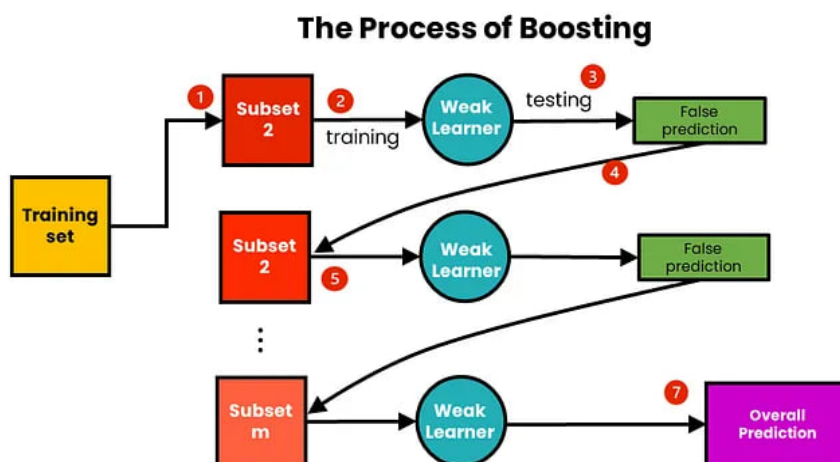


FIGURE 3.4: Boosting algorithm example

Stacking, also known as stacked generalization or meta-ensembling, is an ensemble technique that combines the predictions of multiple base models using a meta-model, i.e. meta-model learns to make a final prediction based on the predictions of the learner models. It goes beyond bagging and boosting by leveraging the strengths of different types of algorithms rather than just multiple instances of the same algorithm.

It enables the model to learn how to combine predictions given by learner models with meta-models and prepare a final model with accurate prediction

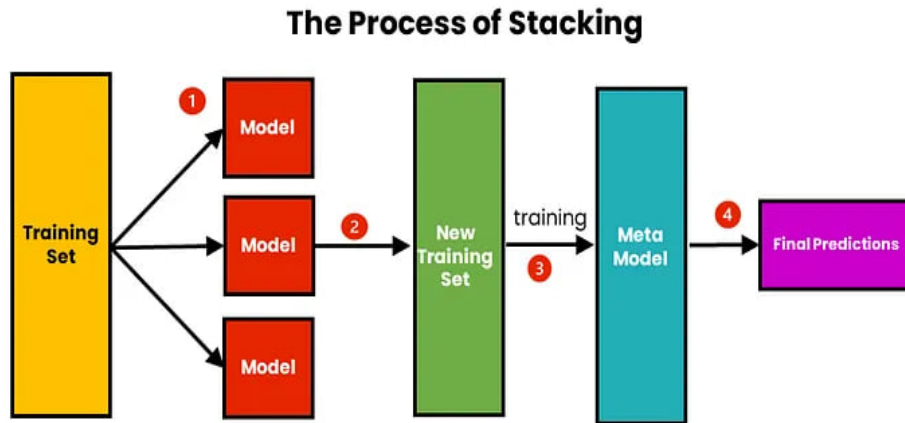


FIGURE 3.5: Stacking algorithm example

Before building the model, we may perform cross-validation with hyperparameter tuning to find out the optimized hyperparameters for the defined model and approximate results to decide whether the model is suitable or not.

Widely recommended and applied practice is to divide a dataset into training, validation, and test sets. The training set is used for model training, the validation set is used to tune hyperparameters, and the test set is kept separate for final evaluation.

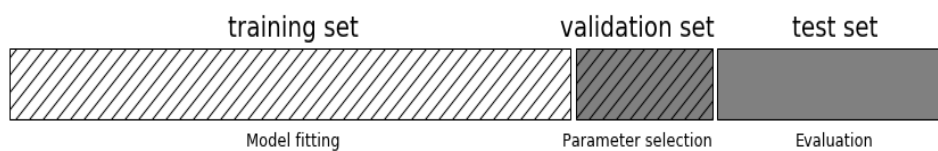


FIGURE 3.6: Threefold split example

However when cross-validating we can use the entire dataset. The data will be split into multiple subsets (folds), training the model on different combinations of these folds, and evaluating the model's performance on the remaining data. This step helps to assess how well the model generalizes to unseen data.

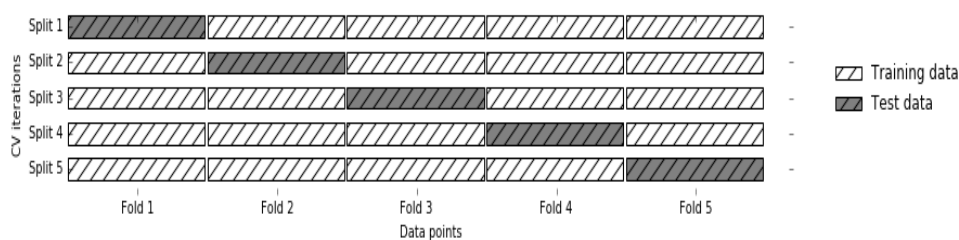


FIGURE 3.7: Cross validation algorithm example

Hyperparameter tuning is the process of finding the best set of hyperparameters for a machine-learning model. For each iteration of cross-validation, a set of hyperparameters is chosen. These hyperparameters could be things like learning rate, number of estimators, maximum depth of the tree, and minimum number of samples to form a leaf node. etc. There are several methods for hyperparameter tuning, including grid search, random search, and Bayesian optimization. In this thesis, we are focusing on grid search which involves defining a grid of hyperparameter values and exhaustively searching through the grid to find the combination that yields the best performance, as assessed by cross-validation.

After performing cross-validation with hyperparameter tuning, we can build our model and evaluate results using corresponding metrics. After trying different models we use evaluation metrics to choose the best one among others.

3.5 Evaluation metrics

Evaluation metrics are quantitative measures used to evaluate the performance of a statistical or ML model. Below are commonly used evaluation metrics, where y is the actual value, $\hat{y}$ is the predicted value and N is the total number of values:

Mean Absolute Error (MAE): The average of the absolute differences between predicted and actual values.

$$\text{MAE}(y, \hat{y}) = \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{N}$$

Root Mean Squared Error (RMSE): The square root of the MSE, provides an interpretable measure in the same units as the target variable.

$$\text{RMSE}(y, \hat{y}) = \sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N}}$$

Symmetric Mean Absolute Percentage Error (SMAPE): The percentage difference between the predicted and actual values, considering the scale of the values being compared. Its values can range from 0% to 200%, where a lower value indicates better model performance and a higher value indicates a larger discrepancy between the actual and predicted values. Although it is commonly used to evaluate the accuracy of forecast models, it can be used with other models to gain a new perspective on the research.

$$\text{SMAPE}(y, \hat{y}) = \frac{100\%}{N} \sum_{i=0}^{N-1} \frac{2 * |y_i - \hat{y}_i|}{|y_i| + |\hat{y}_i|}$$

The evaluation metrics discussed in this chapter will help us determine the performance of predictive models in our research, as they serve as important benchmarks for comparing different models and guiding further research efforts.

Chapter 4

Qualitative analysis

4.1 Conducting Interviews

My professional network allows me to conduct interviews as part of qualitative analysis. I conducted many interviews with different people with diverse backgrounds and skills such as Maksym Mykal, co-founder of LenGreo (digital marketing agency) Viktor Shumylo, CEO of CodeSquad, (mobile, web, and software development agency) with total earnings of \$10K+ and \$100K+ accordingly. I chose these interviewees for their extensive experience in the world of freelancing. To gain a deeper understanding of the data, I created a series of questions covering all aspects of the study. These questions will shed light on areas that I did not fully understand.

Below is a list of the questions with corresponding conclusions:

1. Is it worth considering other factors besides hourly rate or estimated budget before applying for a job and why?

Yes, all available factors should be considered. Employers may use placeholders or omit mentioning pricing information, hence attention must be given to multiple factors like client location, feedback ratings, and the project description.

2. If employers would not specify any budget details, how would you approach finding projects?

If employers would not specify any budget details, a manual approach should be adopted to ensure no potential jobs aligning with the skill set are missed because filters and Upwork suggestions do not have the capacity to weigh all the necessary factors and make the right decision

3. What signs in the job post raise concerns for you, such as low rates, client location, or ratings?

One of the most serious warning signs is the potential for communication and payment issues if the client is located in a third-world country. A significant mismatch between high expectations and low rates can raise concerns about project quality, especially if they are below the market average. Similarly, low client ratings and lots of feedback mean that the client may be incompetent and conflicts may arise.

4. Which hourly and fixed-price rates do you find suspiciously low or high and why?

Identifying suspiciously low or high hourly and fixed-price rates on Upwork can be really tricky. Some people abuse the platform by posting fake jobs for fraudulent purposes, self-promotion, or to gather information about other freelancers' cover letters and portfolios. Determining the authenticity of a job just by looking at its budget is wrong because every expertise level requires appropriate payment, which can vary widely.

While hourly rates as low as \$3 per hour and fixed prices of \$5 are acceptable for certain tasks, high budgets, such as \$200 per hour or \$5000 fixed price, raise concerns, as employers often refrain from placing overly generous budgets to avoid

attracting freelancers motivated by money rather than job quality. But most jobs are priced below \$70 per hour and \$1000 fixed price.

5. What job description/title elements affect your decision to apply and how do you prioritize them? (Length, language complexity, grammar)

While grammar and length are important, priority is given to the overall fit for skills and experience. Even if the description contains minor grammatical errors or is shorter than average, it is inappropriate to reject an application if other factors are good.

6. What categorical factors affect your decision to apply and how do you prioritize them? (Country, experience level desired, payment situations, enterprise client)

Client country holds significant importance because it can influence communication and work dynamics. The most wanted clients are typically found in the USA and throughout Europe. The desired experience level by the client is not considered, as it may not always align with job requirements due to the employer's ignorance. While verified payment is preferred, unverified payment methods are not necessarily bad, as some employers may be new to the platform. Although enterprise clients are desirable, they are extremely rare on Upwork, as large corporations often have alternative methods for recruiting. Accordingly, many freelancers apply, but only the most qualified candidates will be hired.

7. What numerical factors affect your decision to apply, besides the project's budget, and how do you prioritize them? (Rating, feedback amount, total spent)

The number of feedback holds a higher significance as it provides insights from multiple perspectives about the employer. However, feedback absence is not necessarily bad, as it could simply indicate that the employer is new to the platform. Similarly, having a high total spent is a positive thing, but not having it is not necessarily a negative thing.

4.2 Exploring Related Studies

4.2.1 Upwork Jobs Analysis by Z. Faisal

In this research, Z. Faisal worked with "Upwork Job Postings Dataset 2024 (50K Records)" made by A. Rathnayake.

Firstly, he loaded the dataset and examined its contents by checking the dimensions and first rows. Following this, he built a box plot and a density plot. After, he got rid of the outliers by calculating IQR and deleting values less than minimum and more than maximum. Below are showcased initial plots.

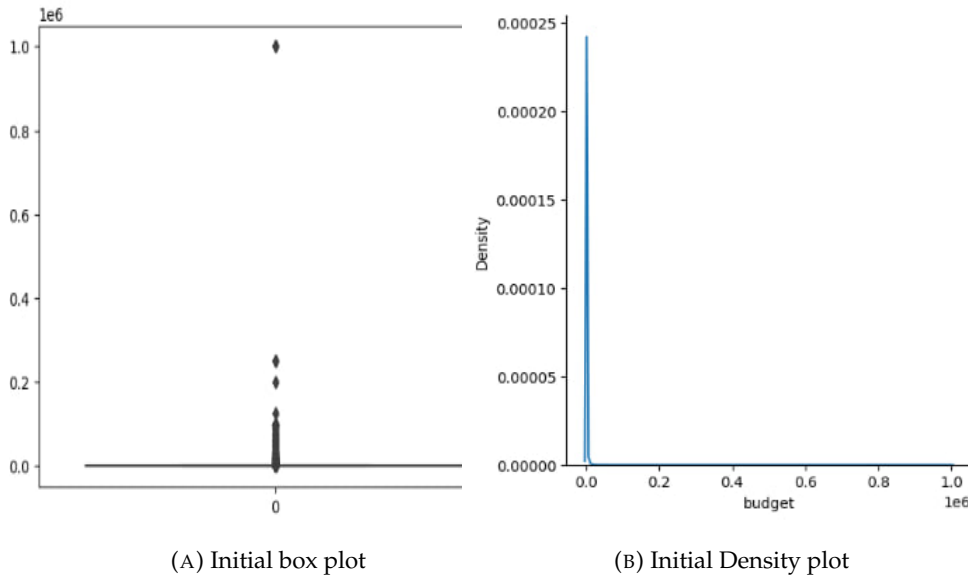


FIGURE 4.1: Initial box and density plots

And here normalized plots are shown:

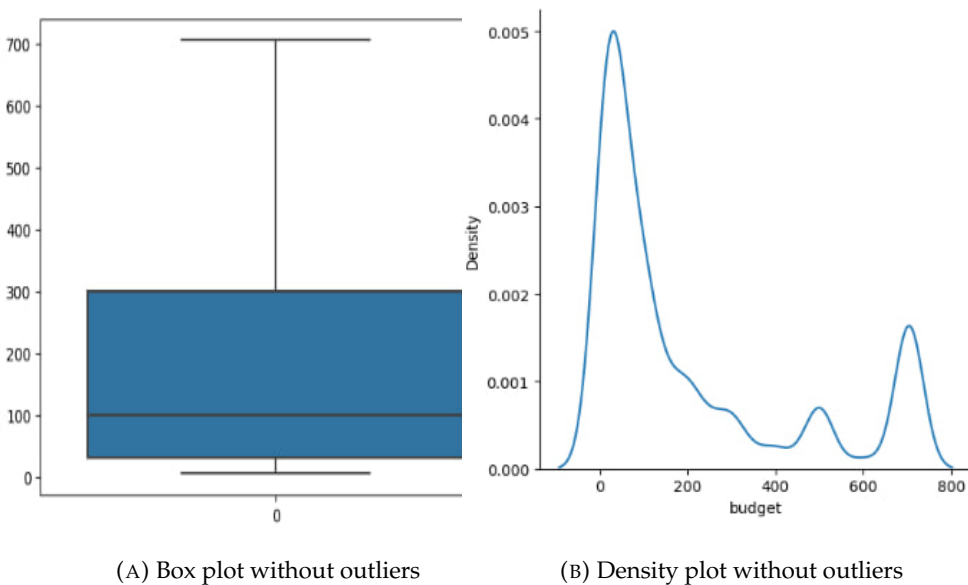


FIGURE 4.2: Box and density plots without outliers

We can see three defined clusters: 1 big one, 1 medium, and 1 small cluster, so I decided to apply an elbow method for determining the optimized number of clusters, as It was not included in the research.

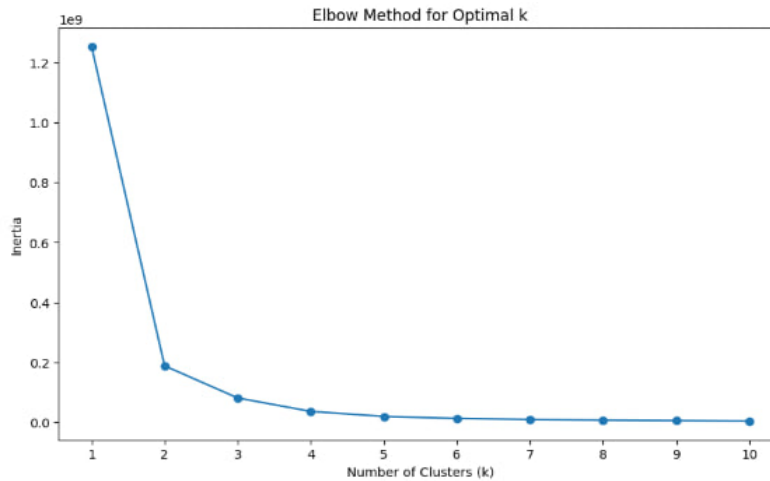


FIGURE 4.3: Elbow method for an optimized number of clusters.

Based on the Elbow method graphic we can see that the optimal number of clusters is 2. We can assume that the same situation will be with our dataset.

Moving on, Zain performed a geographical analysis to learn which countries have the highest number of entries in the dataset.

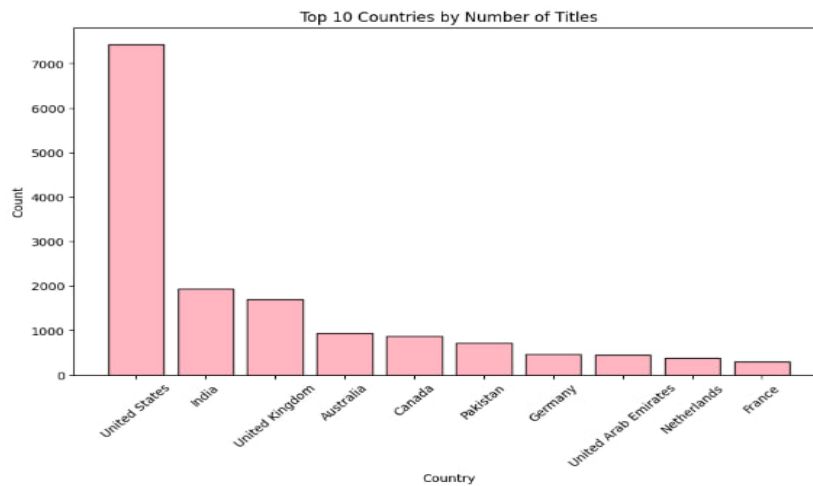


FIGURE 4.4: Top 10 Countries by Number of Titles.

Finally, Zain ventured into predictive modeling, preprocessing the data by handling missing values. For numeric variables such as `hourly_low` and `hourly_high` (start and end of the range accordingly), he applied mean amputation; for nominal categorized variables such as `title` and `country` he performed one-hot encoding with mode imputation, for ordinal categorized variable `is_hourly` he performed nominal encoding with mode imputation.

For regression he chose a linear one and here is the pipeline of his model:

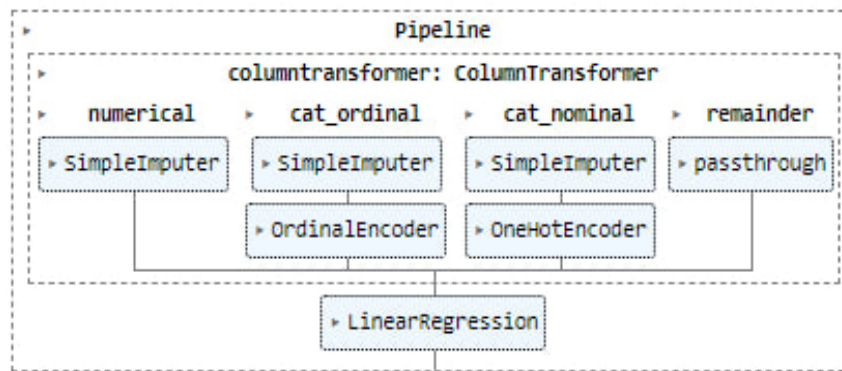


FIGURE 4.5: Model pipeline

Then he trained a linear regression model to predict budgets based on features, evaluating its performance using metrics such as RMSE and R^2 score for both training and testing datasets.

Training RMSE: 37.8

Testing RMSE: 155.4

Training R^2 score: 0.9

Testing R^2 score: -0.03

From these calculations, we can confirm that the model's performance is bad due to the model over-fitting the training set.

4.2.2 Hourly wages Upwork CatBoost+SHAP by D. Iakubovskiy

In this research, Dmytro Iakubovskiy, Senior Data Scientist at Velux, worked with "All Upwork Job Postings - Monthly Tracker" dataset made by A. Rathnayake.

Firstly, he loaded the data and preprocessed it by deleting duplicates, removing titles longer than 50 characters, and choosing the columns he wanted to work with, namely 'hourly_avg', 'title', and 'country' columns.

Then, the data scientist proceeded to feature engineering by setting up the rare label encoder limiting the number of titles to 69 and countries to 91.

Unlike Z. Faisal, Dmytro performed ML by dividing the dataset into two equal training and testing sets and choosing a tree-based model with ensemble method boosting, CatBoostRegressor, and configuring it with the next parameters:

```

depth=6,
verbose=0,
early_stopping_rounds=100,
learning_rate=0.008,
loss_function='RMSE'

```

And achieved the following results:

RMSE score for train: 31.151 USD/hour

RMSE score for test: 30.113 USD/hour

RMSE baseline score for train: 31.443 USD/hour

RMSE baseline score for test: 30.373 USD/hour

In addition to RMSE, Dmytro used SHapley Additive exPlanations (SHAP) values to get a consistent and objective explanation of how each feature impacts the model’s prediction: evaluation:

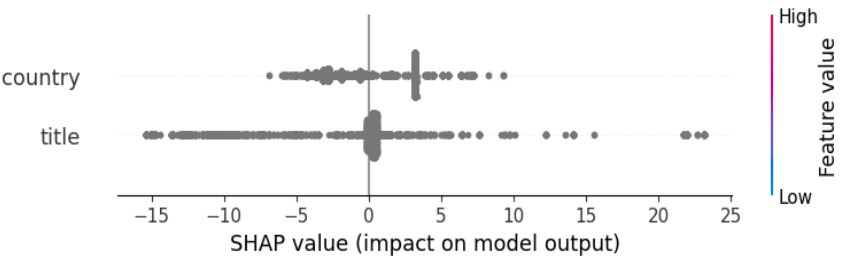


FIGURE 4.6: SHAP value

Then he visualized SHAP values for countries and titles individually:

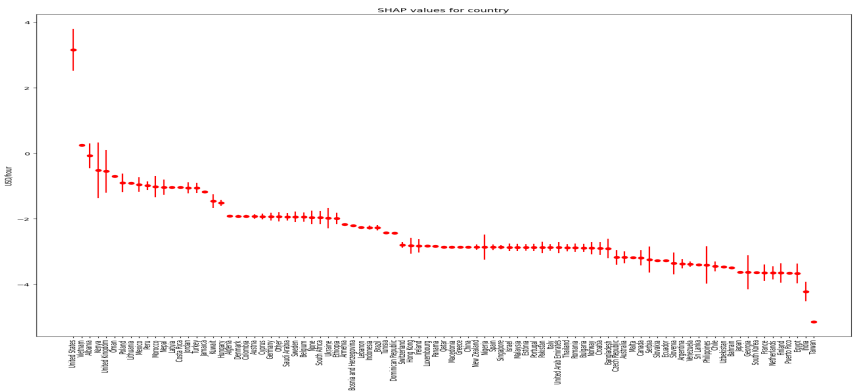


FIGURE 4.7: SHAP country

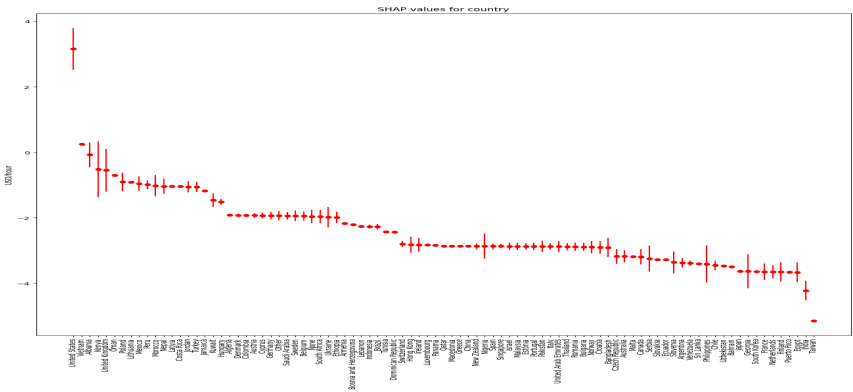


FIGURE 4.8: SHAP title

These graphs allow us to evaluate how each feature impacted the model’s prediction.

In conclusion, Z. Faisal and D. Iakubovskiy conducted two comprehensive studies that involved data exploration, trend analysis, visualization, predictive modeling, and performance evaluation. These studies provided me with a new perspective on my dataset.

4.3 Hypotheses Formation

After conducting interviews and exploring related studies we are confident in forming next hypotheses:

1. There is little to no significant correlation between separate factors and project budget, but the country will show the best significance.
2. Linear regressions are not suitable to show these relationships, tree-based models will show better results.
3. The more factors are involved in the model, the better the result will be.
4. The optimal number of clusters in my data will be 2.
5. For fixed-price projects, outliers will be prices exceeding \$1000, and for hourly-rated projects, those surpassing \$75.

Chapter 5

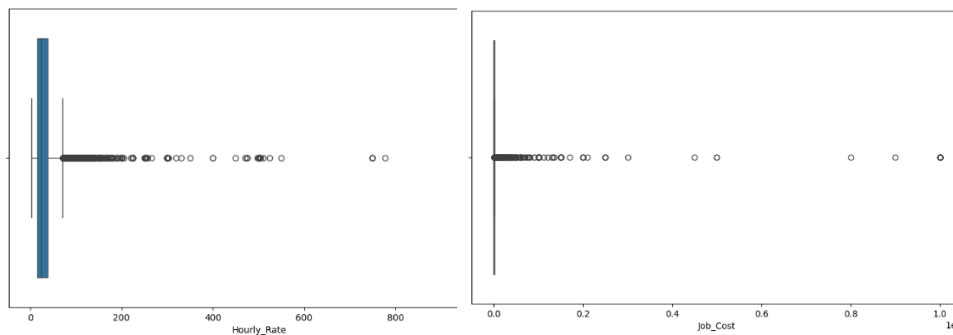
Quantitative analysis

5.1 Preprocessing of hourly_rate and job_cost columns

To begin with, we should split our dataset into two datasets: one containing projects that are hourly-based and the other with the ones that are fixed-price. Additionally, we need to convert the Hourly_Rate and Job_Cost columns from String to Float. We also need to preprocess the Hourly_Rate column by calculating the average value as some values are expressed in a range format with both low and high values.

Then we need to remove any duplicate data entries and fill in any empty values to ensure the accuracy and consistency of the data. For numeric columns, we will replace empty values with 0, while for categorized columns, we will replace empty values with the mode (most frequently occurring value). In the case of the Enterprise_Client column, empty values will be replaced with "NOT_Enterprise" values, as only Enterprise clients are marked in this column.

Next, it is crucial to see the data distribution by creating a boxplot. This will help us gain a better understanding of the data we are working with.



(A) Initial box plot for hourly-rated jobs (B) Initial box plot for fixed-price jobs

FIGURE 5.1: Initial box plots

Based on the boxplots, it is evident that the dataset contains numerous outliers. To address this issue, we need to define the IQR and eliminate any values that are more than 1.5 times the IQR below the first quartile data or above the third quartile. From these calculations, we see values greater than \$71.25 for hourly-based projects and \$1175.0 for fixed-price projects are considered outliers and it proves that our Hypothesis 5 was close to the truth.

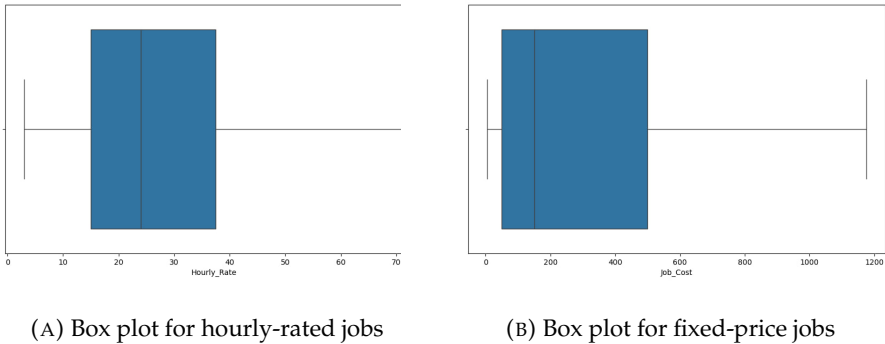


FIGURE 5.2: Box plots without outliers

Now we have to build a density plot for our fixed-price values to verify or refute our Hypothesis 4.

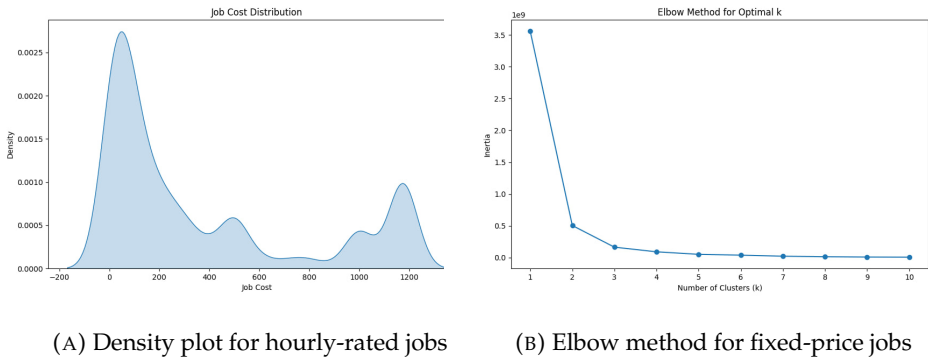


FIGURE 5.3: Density plot and Elbow method

From these plots we observe that the elbow method plot for the optimized number of clusters is nearly identical, despite variations in distribution and range, verifying our Hypothesis 4.

As all subsequent analyses resulted in similar outcomes for hourly-based and fixed-price projects, only fixed-price projects will be described.

5.2 Categorical Variables

First, we need to calculate the count, mean, median, mode, minimum, and maximum values for each category to determine if there are any variations between categories.

Client Country	min	max	count	mean	median	mode
United States	5.0	1175.0	6222	446.9	250.0	100.0
India	5.0	1175.0	2854	290.9	100.0	100.0
United Kingdom	5.0	1175.0	1478	364.9	150.0	50.0
Canada	5.0	1175.0	939	378.2	200.0	100.0
Pakistan	5.0	1175.0	849	155.6	35.0	10.0
Australia	5.0	1175.0	761	370.8	200.0	100.0
Germany	5.0	1175.0	427	365.0	150.0	100.0
United Arab Emirates	5.0	1175.0	416	363.1	150.0	100.0
Romania	5.0	1175.0	283	436.4	200.0	1000.0
Netherlands	5.0	1175.0	250	406.5	200.0	50.0

TABLE 5.1: Statistics for the 10 most popular countries with job cost (sorted by count)

EX_level_demand	min	max	count	mean	median	mode
Intermediate	5.0	1175.0	8567	356.9	150.0	100.0
Expert	5.0	1175.0	6554	417.3	200.0	100.0
Entry level	5.0	1175	2059	272.8	75.0	100.0

TABLE 5.2: Statistics for Expert Level Demanded with job cost (sorted by count)

Search_Keyword	min	max	count	mean	median	mode
Developer	5.0	1175.0	12121	382.9	150.0	100.0
Marketing	5.0	1175.0	6325	362.9	200.0	100.0
Data_science	5.0	1175.0	1667	331.0	100.0	100.0

TABLE 5.3: Statistics for top 3 Search Keywords with job cost (sorted by count)

Payment_Situation	min	max	count	mean	median	mode
Payment verified	5.0	1175.0	14002	341.1	125.0	100.0
Payment unverified	5.0	1175.0	6111	443.4	250.0	100.0

TABLE 5.4: Statistics for Payment Situation with job cost with job cost (sorted by count)

Enterprise_Client	min	max	count	mean	median	mode
NOT Enterprise	5.0	1175.0	20101	371.9	150.0	100.0
Enterprise	50.0	1175.0	12	811.7	1000.0	1000.0

TABLE 5.5: Statistics for Enterprise clients with job cost with job cost (sorted by count)

Based on these tables, it is evident that the categories Enterprise_Client and Client_Country exhibit the largest difference between their minimum and maximum mean values. For Enterprise_Client, the difference is 439.8 (811.7 - 371.9), whereas for Client_Country, it is 291.3 (446.9 - 155.6). This indicates that these metrics can be useful in selecting potential job opportunities. It also verifies the second part of hypothesis 1.

After encoding 'Search_Keyword', 'Client_Country', 'Payment_Situation' with nominal encoder and 'EX_level_demand' with ordinal encoder respectfully, we can build a heatmap to explore the strength and direction of correlation of these characteristics with Job_Cost.

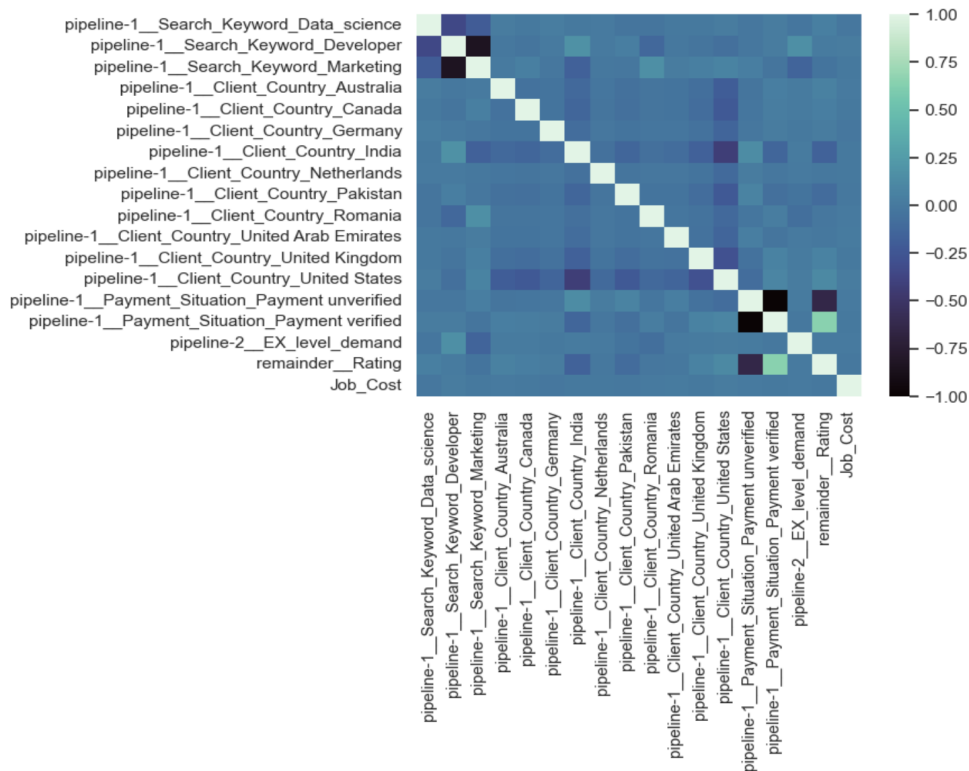


FIGURE 5.4: Heatmap for categorical variables

This heat map verifies the first part of hypothesis 1 saying about low correction between separate factors and the project budget.

5.3 Numeric Variables

For numeric variables, we have three columns in the dataset - Spent(\$), Feedback_Num, and Rating - which also have a significant number of outliers. Using the IQR method we will delete values greater than 5000.0 for Spent(\$), and 27.5 for Feedback_Num to normalize the data.

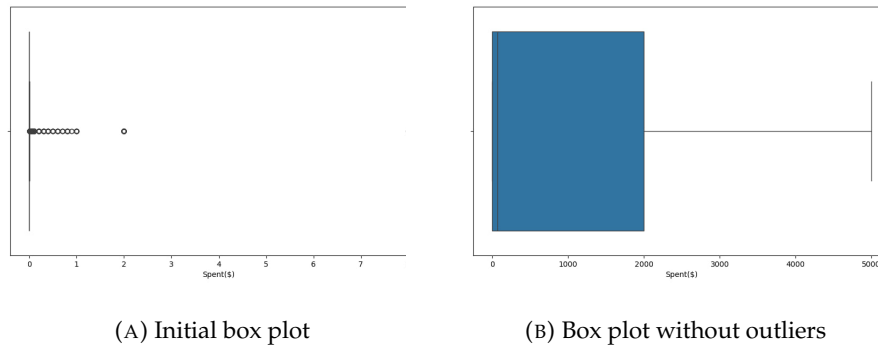


FIGURE 5.5: Cleansing Spent(\$) column

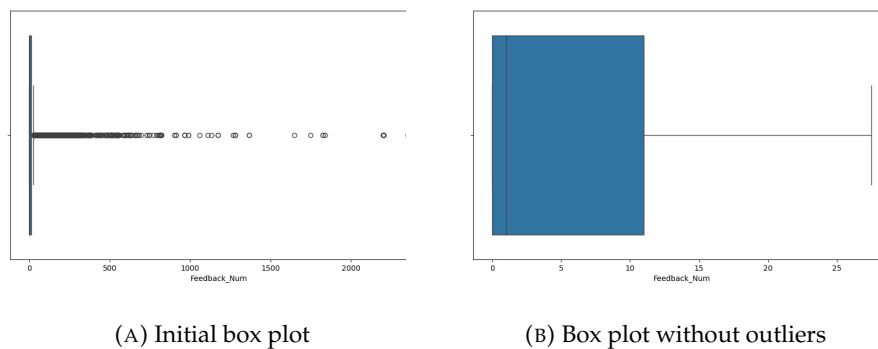


FIGURE 5.6: Cleansing Feedback_Num column

Then we can proceed to creating a heatmap to explore the strength and direction of the correlation of our numeric factors with Job_Cost.

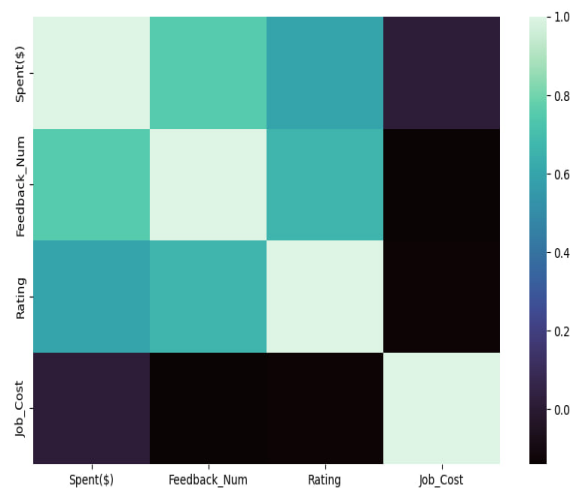


FIGURE 5.7: Heatmap for numeric variables

This heatmap shows that our numerical features have almost no correlation with Job_Cost and also verifies the first part of hypothesis 1 saying about no correction between separate factors and the project budget.

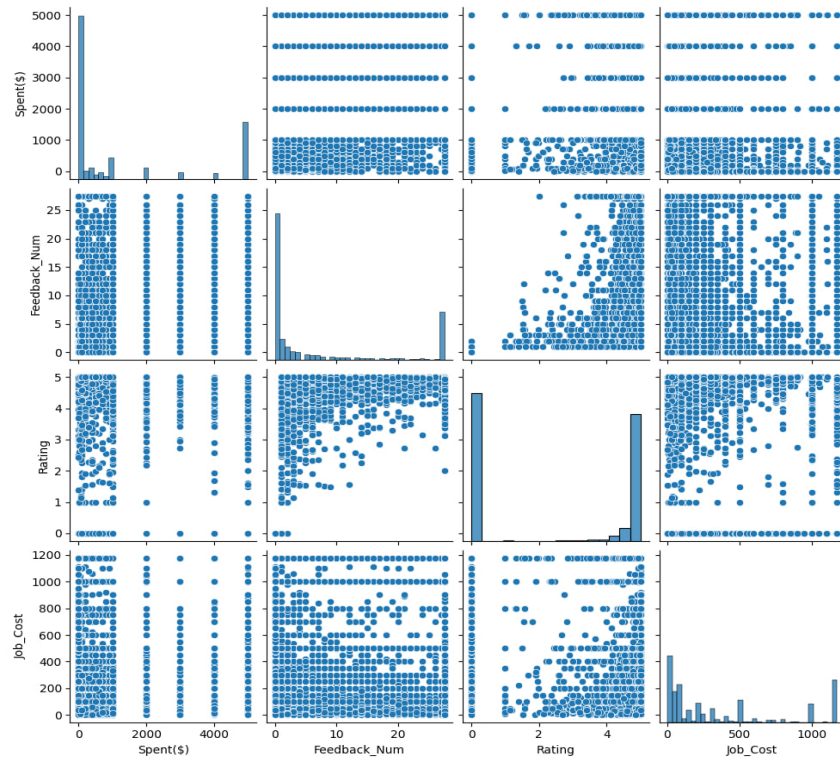


FIGURE 5.8: Pairplot for numeric variables

Chapter 6

Prediction Models: Results

After preprocessing our data and conducting correlation analysis, we will experiment with various models to identify those exhibiting the worst and best performance.

6.1 Categorical Variables

Worst results: Linear regression with default hyperparameters.

RMSE: 279.1
MAE: 213.5
SMAPE: 99.0

Best results: Gradient boosting regression with tailored hyperparameters.

RMSE: 293.0
MAE: 184.1
SMAPE: 91.4

6.2 Numeric Variables

Worst results: Linear regression with default hyperparameters

RMSE: 319.9
MAE: 199.3
SMAPE: 95.9

Best results: Gradient boosting regression with tailored hyperparameters.

RMSE: 299.9
MAE: 195.9
SMAPE: 92.4

6.3 Text Variables

Note: Cross-validating with NLP is computationally intensive and demands the application of a preprocessing pipeline on each fold separately. Given that we achieved optimal results with the selected hyperparameters, we prioritized utilizing our computational resources to cross-validate the entire preprocessing pipeline with a stacked model. This stacked model encompasses textual, numeric, and categorical features

Worst results: Gradient boosting regression.

RMSE: 234.9
MAE: 167.4
SMAPE: 91.9

Best results: XGB regression.

RMSE: 233.4
MAE: 164.2
SMAPE: 90.0

6.4 Complex Model

Worst results: Linear regression with default hyperparameters.

RMSE: 289.0
MAE: 177.5
SMAPE: 83.1

Best results: XGB Regression with default hyperparameters.

RMSE: 297.8
MAE: 184.8
SMAPE: 79.4

These results confirm hypotheses 3 and 2, even though the best tree-based models show results not much superior to linear ones.

Chapter 7

Conclusions

7.1 Summary

In this thesis, we analyzed the impact of diverse aspects on a project's budget to study how to maximize a freelancer's profit on the Upwork platform. We used quantitative, qualitative, NLP analyses and ML to demonstrate what these dependencies look like and discovered that it is impossible to determine the profitability of the project based on separate factors because they do not have a strong correlation with the budget individually. We proved that it is necessary to comprehensively analyze all factors and nuances in order to grasp a holistic understanding, beyond the scope of mere numbers and algorithms.

7.2 Further work

Explore another way of maximizing freelancer's profit on Upwork by optimizing freelancer profiles and writing cover letters with relevant portfolios attached.

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